

# HOSPITAL MANAGEMENT SYSTEM

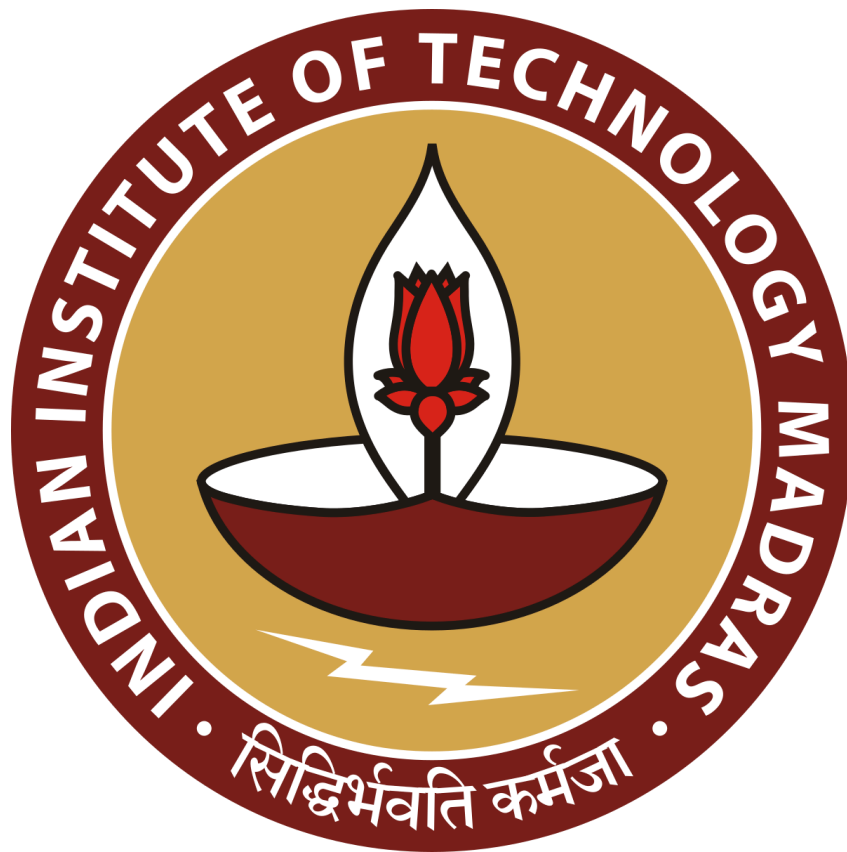
## MAD-1 PROJECT REPORT

Submitted by

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## Project Details

### Project Title: Hospital Management System

The Hospital Management System (ApexCare) is designed to streamline the operational flows of a modern healthcare facility. Developed as part of IITM BS Degree's Modern Application Development I (MAD-I) course, the application facilitates seamless interaction between three key people: Administrators, Doctors, and Patients.

The system replaces manual record-keeping with a centralized database, ensuring data integrity, preventing appointment conflicts, and providing role-specific dashboards for efficient management.

## Technologies Used

- **Backend** : Python Flask
- **Database**: SQLite (Relational Database)
- **ORM**: Flask-SQLAlchemy
- **Frontend**: HTML5, CSS3, Bootstrap 5 (Responsive UI), Jinja2 Templating
- **Forms & Validation**: Flask-WTF
- **Version Control**: Git

## AI/LLM Declaration

I used **Google Gemini** to assist in writing HTML Templates and the dummy headings and paragraphs on the landing page. The extent of AI/LLM usage is around **5–10%**, limited to **code suggestions and documentation formatting**.

All final implementation logic, debugging, and integration were done manually.

# Features

## 1. Core Functionalities

- **Authentication:** Secure login for all roles and registration for patients.
- **Session Management:** Secure server-side session handling to protect routes.
- **Conflict Prevention:** Logic to prevent double-booking of doctors for the same time slot.

## 2. Admin Module

- **User Management:** Ability to Add, Edit, Blacklist, or Remove Doctors and Patients.
- **One-Time Credential View:** Secure generation of doctor credentials with a single-view policy for enhanced security.
- **Appointment Oversight:** View all hospital appointments and detailed treatment records.

## 3. Doctor Module

- **Dashboard:** Tabs for "Upcoming Appointments" and "Assigned Patients".
- **Availability Management:** Interface to update availability for morning/evening slots for the next 7 days.
- **Treatment Management:** Functionality to mark appointments as "Completed" or "Cancelled" and enter Diagnosis/Prescription details.
- **Patient History:** Access to the full medical history of assigned patients.

## 4. Patient Module

- **Department Browsing:** View department details and lists of specialists.
- **Doctor Search:** Search functionality for doctors by name or specialization.
- **Appointment Booking:** Dynamic booking system based on real-time doctor availability.
- **Management:** Options to Reschedule or Cancel booked appointments.
- **Medical History:** View past appointments, including diagnoses and prescriptions.

## 5. API Module

- **RESTful Architecture:** Implemented using the Flask-Restful extension to provide programmatic access to hospital data.
- **Resource Endpoints:** Dedicated GET endpoints for Doctors, Patients, and Appointments.
- **JSON Serialization:** Returns data in standard JSON format, enabling potential integration with external mobile apps or third-party services.

## Database Schema / ER Diagram

